

Control Panel For SUBMERSIBLE MONOBLOCK PUMPSET

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Presented here is a submersible pump starter circuit using electronic overload relay, solid-state relay and adjustable startup delay.

A submersible pump is a type of centrifugal pump designed to function with the pump and the motor completely submerged in the water. The motor is sealed in such a way that it prevents any ingress of water from seeping into the motor casing and causing it to fail.

Since the pump is submerged in water, floor space can often be better utilised. Overall cost of the installation can also be reduced. This type of pump tends to have lower maintenance cost and creates less noise than a pump installed above ground.

Capacitor-start and capacitor-run motors are widely used in submersible pumps. These are used in a wide range of single-phase applications primarily for starting hard loads. These motors have moderate-to-high starting torque compared to other types of single-phase motors. Starting torque generally ranges from 200 to 350 per cent of normal full-load torque.

The main advantage of a capacitor-start motor is the lower starting torque, which results in lower starting current. This motor is generally used for single-phase applications between three and 10 horsepower.

Since the pump is submerged, there are no issues with pump priming either. But fixing a centrifugal switch and capacitor inside the motor can be dangerous. So, capacitors are fixed inside the control panel. The most complicated part in this system is replacing the inbuilt mechanical centrifugal switch with an external electronic centrifugal switch.

In the older version of the starter, centrifugal cut-off is achieved manually by pressing and releasing a pushbutton switch. But in the latest version, automatic switch operations

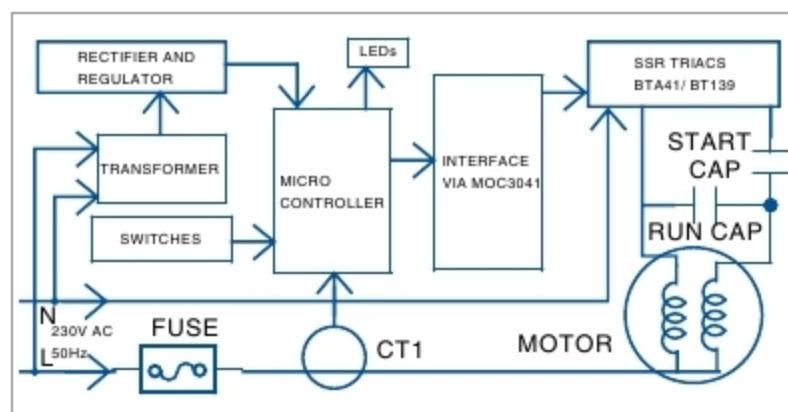


Fig. 1: Block diagram of the project

TIME DELAY SETTINGS USING DIP SWITCH

DIP Switch Position	Delay	P3.5	P3.4	P3.3
(OFF, OFF, OFF)	5.0 seconds	1	1	1
(OFF, OFF, ON)	4.5 seconds	1	1	0
(OFF, ON, OFF)	4.0 seconds	1	0	1
(OFF, ON, ON)	3.5 seconds	1	0	0
(ON, OFF, OFF)	3.0 seconds	0	1	1
(ON, OFF, ON)	2.5 seconds	0	1	0
(ON, ON, OFF)	2.0 seconds	0	0	1
(ON, ON, ON)	1.5 seconds	0	0	0

and delays are provided. The control panel should be a foolproof system to protect the motor from any damage.

Main disadvantages of starters used in older submersible pumps are as follows:

1. Electromechanical relays employed have a limited operational life.
2. Mechanical overload trip mechanisms have less sensitivity.
3. Startup time delay cannot be changed.

This project solves the problems of existing submersible motor starters by using an electronic overload relay, a solid-state relay in place of an electromechanical relay and an adjustable startup delay. Fig. 1 shows the block diagram of the project.

Circuit and working

Fig. 2 shows the circuit diagram of the submersible motor control panel.

The circuit is built using 7805 regulator (IC1), microcontroller (MCU) IC AT89C4051 (IC2), two BT139/BTA41 triacs, two triac driver ICs MOC 3041 (IC3 and IC4), three transistors (T1 through T3), 10A, 50Hz current transformer (CT1) and a few discrete components.

The circuit is powered with 230V AC mains. Transformer X1 steps down voltage to 9V-0-9V. Diodes D1 and D2 act as rectifier diodes, and capacitors C1 and C2 are connected as filters. Rectified DC voltage is regulated to 5V using IC1 and fed to MCU circuit.

The heart of the circuit is the MCU IC AT89C4051 running at an oscillator frequency of 12MHz, by connecting crystal XTAL1. Power triacs BT139 are interfaced to the MCU through triac driver ICs and transistors.

AT89C4051. AT89C4051 is a low-voltage, high-performance CMOS 8-bit microcomputer with 4kB flash programmable and erasable read-only memory (PEROM). The device is manufactured using Atmel's high-density, non-volatile memory technology. It is compatible with industry-standard MCS-51 instruction set.

By combining a versatile 8-bit CPU with flash on a monolithic chip, AT89C4051 becomes a powerful microcomputer that provides a highly-flexible and cost-effective solution to many embedded control applications.

For details on AT89C4051 MCU, refer to its datasheet on their website, as follows: